

Stock Market Analysis in SQL - Insights

Six NSE stocks (Bajaj Auto, Eicher Motors, Hero Motocorp, Infosys, TCS, TVS Motors), 889 trading days, 1 Jan 2015 - 31 Jul 2018. Strategy studied: 20-day / 50-day moving-average 'golden cross' (Buy when MA20 crosses above MA50, Sell when it crosses below). All numbers below come from the SQL files in this project; the checkpoints from the student guide all pass (see outputs/results/99_checkpoints.csv).

1. Data cleaning and the two price events

Cleaning: dates converted from '31-July-2018' to ISO dates, columns renamed to snake_case, rows sorted oldest-first, no duplicate dates, 6 tables x 889 rows with identical dates (so the master-table join loses nothing). Only gap: *deliverable_qty* and *pct_deli_qty* are blank on **2015-12-09** (Eicher, Hero, TCS, TVS) and **2017-08-31** (Bajaj, Infosys) - the same two dates across several unrelated companies points to the exchange's reporting, not the companies. Blanks were kept as NULL (never as 0); close prices are complete.

Price event 1 - TCS, 2018-05-31: close falls 50.4% in one day (from 3,517.75 to 1,744.80). **Price event 2 - Infosys, 2015-06-15:** -49.9% in one day (to 991.10). Both are 1:1 bonus issues (shares double, price halves, no value lost), not crashes. I divided every close *before* the event date by 2 (task 13) and re-ran the analysis on adjusted prices. No other stock has a one-day move worse than -9.7%.

2. Results at a glance

Stock	Raw %	Adj. %	Buys	Sells	Last signal (adj.)	Win rate*	Avg trade*	vs MA50
Bajaj Auto	+10.0%	+10.0%	12	11	<bound method NDFrame.last of raw adj 10.04849 buys 12 sells 11 last Buy (2018-06-21) trips 11 win 36.363636 avg 0.440813 comp 2.048358 fast 9 vs50 -5.796447 Name: Bajaj Auto, dtype: object>	36%	+0.4%	-5.8%
Eicher Motors	+82.6%	+82.6%	6	7	<bound method NDFrame.last of raw adj 82.562348 buys 6 sells 7 last Sell (2018-06-06) trips 6 win 100.0 avg 10.153038 comp 76.497617 fast 1 vs50 -3.374185 Name: Eicher Motors, dtype: object>	100%	+10.2%	-3.4%

Stock	Raw %	Adj. %	Buys	Sells	Last signal (adj.)	Win rate*	Avg trade*	vs MA50
Hero Motocorp	+6.0%	+6.0%	9	9	<bound method NDFrame.last of raw adj 6.001995 buys 9 sells 9 last Sell (2018-05-22) trips 9 win 33.333333 avg -1.719546 comp -16.339757 fast 3 vs50 -6.210073 Name: Hero Motocorp, dtype: object>	33%	-1.7%	-6.2%
Infosys	-30.9%	+38.2%	10	10	<bound method NDFrame.last of raw adj 38.17188 buys 10 sells 10 last Buy (2018-05-07) trips 9 win 33.333333 avg -0.368657 comp -7.116419 fast 4 vs50 6.180895 Name: Infosys, dtype: object>	33%	-0.4%	+6.2%
TCS	-23.8%	+52.4%	12	12	<bound method NDFrame.last of raw adj 52.362452 buys 12 sells 12 last Buy (2018-04-20) trips 11 win 18.181818 avg -3.354538 comp -31.918915 fast 2 vs50 4.446924 Name: TCS, dtype: object>	18%	-3.4%	+4.4%
TVS Motors	+86.9%	+86.9%	8	8	<bound method NDFrame.last of raw adj 86.906267 buys 8 sells 8 last Sell (2018-05-17) trips 8 win 37.5 avg 11.182226 comp 85.016503 fast 2 vs50 -9.262597 Name: TVS Motors, dtype: object>	38%	+11.2%	-9.3%

*Win rate / average trade: buy at each Buy-day close, sell at the next Sell-day close, ignoring costs and dividends. 'vs MA50' = last close relative to its 50-day average. 'Raw %' is the first-to-last close change on unadjusted prices (task 11); 'Adj. %' uses bonus-adjusted prices (task 13; only TCS and Infosys differ). Signal counts for TCS/Infosys here are on adjusted prices; on raw prices the totals are 56 Buys and 57 Sells across all six stocks (task 10).

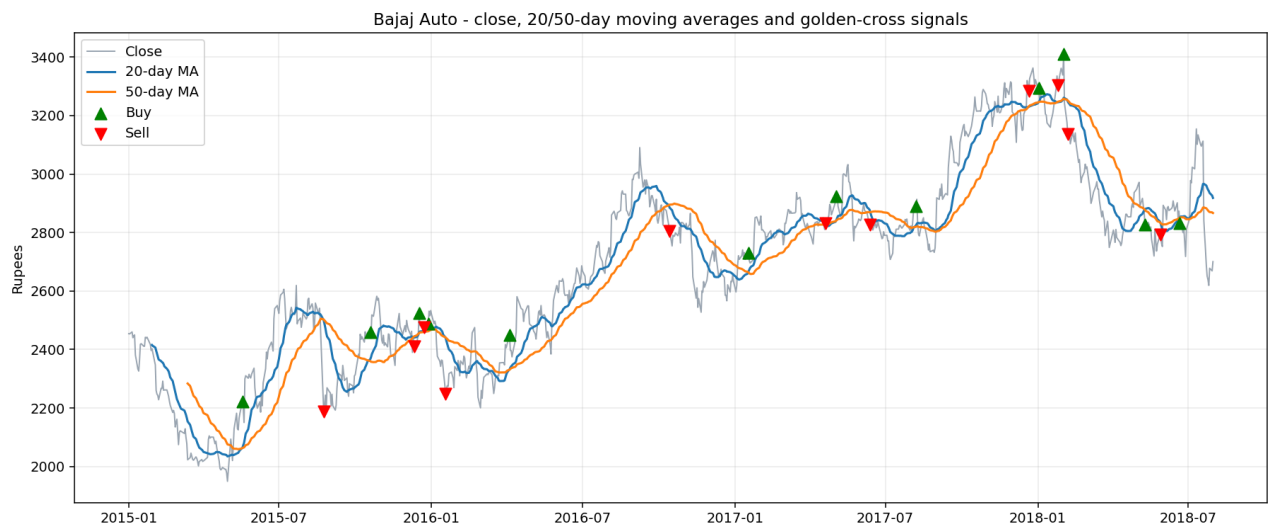


Figure 1: Bajaj Auto with 20/50-day averages; triangles are the golden-cross signals from task 7.

3. Per-stock findings (claim - evidence - caveat)

Claim: TVS Motors and Eicher Motors were the clear winners (+86.9% and +82.6%), and the moving-average rule captured most of it.

Evidence: Task 11/13 first-to-last closes; Eicher produced 6 Buys / 7 Sells with 100% of round trips profitable (compounded +76%); TVS compounded +85%.

Caveat: Only 6-8 round trips each, so a few big trends drive the result. A buy-and-hold investor would have done about as well without any signals. Both end with a Sell signal (Eicher 2018-06-06, TVS 2018-05-17) and TVS closes 9.3% below its 50-day average, but both are still far above their 2015 levels.

Claim: TCS and Infosys went from 'losers' to 'winners' once the bonus issues were handled: TCS -23.8% raw vs +52.4% adjusted; Infosys -30.9% raw vs +38.2% adjusted.

Evidence: Task 11 versus task 13 (outputs/results). Task 12 shows the -50.4% (TCS) and -49.9% (Infosys) single-day moves; Figure 3 shows the cliff removed.

Caveat: Adjustment assumes exactly 1:1 bonus ratios and ignores dividends, so true total return is somewhat higher.

Claim: Bajaj Auto and Hero Motocorp were roughly flat, and the crossover strategy lost money on Hero.

Evidence: Bajaj +10.0%, Hero +6.0% over the full period. Hero: 33% win rate, compounded -16%; Bajaj: 36% win rate, compounded +2%.

Caveat: Sideways markets are exactly where moving-average crossovers whipsaw, so this is a property of the period as much as of the stocks.

Claim: There were clusters of rapid Buy/Sell flips (whipsaws), mainly in Bajaj Auto.

Evidence: Bajaj had 9 signals within 21 days of the previous one. Example: Buy 2015-12-28, Sell 2016-01-18 (-9.6%), and Buy 2018-02-01, Sell 2018-02-06 (-8.0%).

Caveat: Round-trip losses ignore brokerage and taxes, which would make frequent flipping even more expensive.

Claim: The latest signal does not always match the long-run trend.

Evidence: Bajaj's last signal is Buy (2018-06-21) while its last close is -5.8% below its 50-day average; Infosys and TCS (adjusted) both end on Buy and sit above their 50-day averages. On raw prices TCS shows a false Sell on 2018-06-05, created purely by the bonus cliff (task 14: 1 TCS signal removed by adjustment); Infosys has 4 changed signals (1 removed, 3 created) around the 2015 cliff.

Caveat: I would trust the long-run trend and the adjusted series over a single recent crossover: crossings lag the price, and one signal carries little information.

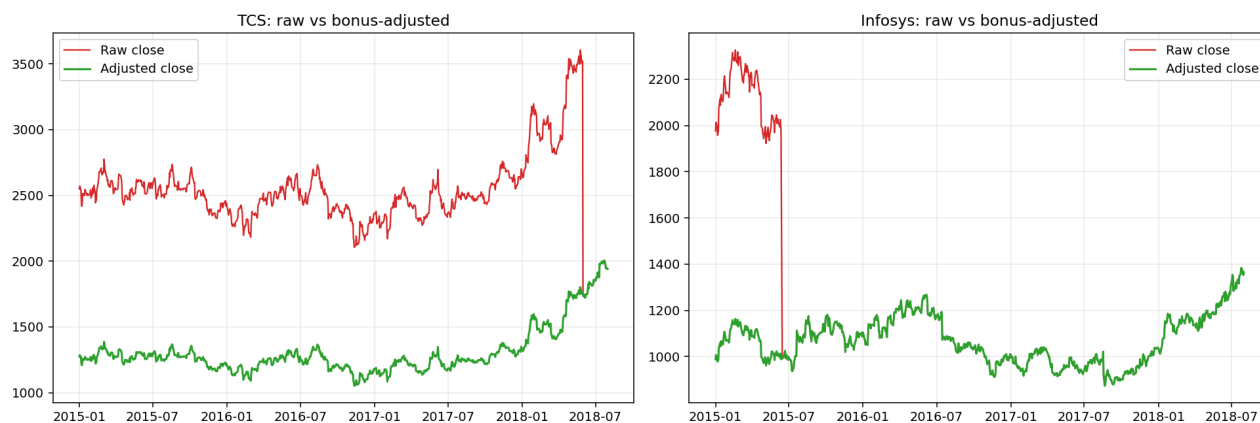


Figure 3: raw vs bonus-adjusted close for TCS and Infosys (red = raw, green = adjusted).

4. Questions about the method

Claim: A golden cross is a lagging signal: it can only confirm a trend that has already moved.

Evidence: The 50-day average needs 50 trading days of history, so the first Bajaj signal was possible only after 2015-03-13; the first actual Buy was 2015-05-18. In every stock the average is made of past prices only.

Caveat: By the time MA20 crosses MA50 a good part of the move has usually happened; the table above shows modest or negative average trades for four of six stocks.

Claim: Several conclusions in the course deck change after fixing the price events.

Evidence: Deck: Infosys '3% decrease' - correct figures are -30.9% raw and +38.2% adjusted. Deck: TCS '23.8% decrease' - raw is correct (-23.8%) but adjusted is +52.4%, and its 'sell' call reverses (adjusted last signal is Buy, 2018-04-20). Deck TCS first close '2454.1' is Bajaj's price (TCS actually 2548.20). Eicher '82.5%' is 82.6% when rounded. Buy/Sell counts for Bajaj (12/11), Eicher (6/7), Hero (9/9), Infosys (9/9), TCS (12/13), TVS (8/8) match the raw-price results.

Caveat: The deck's final 'buy Bajaj and Infosys; sell Eicher, TVS, Hero, TCS' list should not be used as is: it mixes unadjusted TCS/Infosys prices with signals,.

Claim: This data is not enough for a real investment decision.

Evidence: The tables contain only daily prices and volumes for six stocks.

Caveat: Missing: dividends, news and earnings, brokerage/tax costs, slippage, risk measures (volatility, drawdown), comparison with the Nifty index, and any test on data outside 2015-2018.

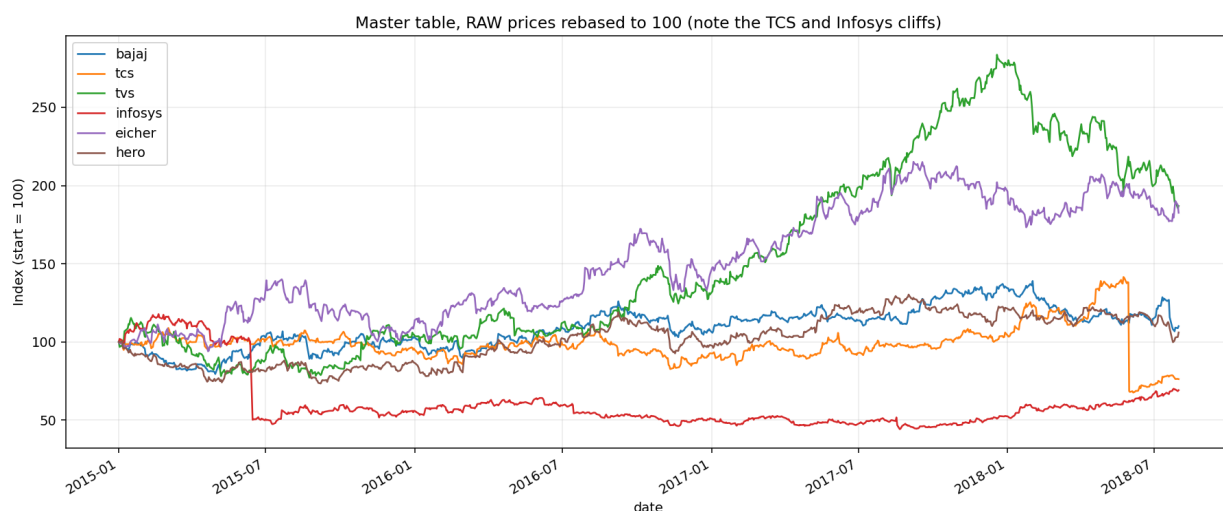


Figure 2: master table, raw closes rebased to 100 - the TCS and Infosys cliffs are visible as vertical drops.